

ADISHESH GONIBEED RAVISHANKAR

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EDUCATION

Arizona State University

Master of Science in **Computer Software Engineering** | GPA: 3.83/4

Thesis: Use of Deep Reinforcement Learning and LLMs for Ontology Alignment

Advisor: Dr. Srividya Bansal

Tempe, AZ

Aug. 2024 – May 2026

Visvesvaraya Technological University

Bachelor of Engineering in **Information Science and Engineering**

Bengaluru, India

Aug. 2023

TECHNICAL SKILLS

Languages: R, Python, C, C++, JavaScript, CSS, SQL, PHP, Java, Scala.

Libraries & Tools: PyTorch, TensorFlow, Keras, Hugging Face, LangChain, PEFT (LoRA/QLoRA), FHE, FAISS, Scikit-learn, Pandas, Matplotlib, OpenCV, Mediapipe, Node.js, React.

Cloud & DevOps: AWS (EC2, S3, Route 53, CloudFront, Developer Tools), Kubernetes, Docker, Git, Agile, Tableau, Spark, Kafka.

Certifications: Generative AI for Everyone (DeepLearning.ai), AWS Cloud Practitioner, RCR - Responsible Conduct of Research, IRB - Social, Behavioral Research (CITI Program).

RESEARCH EXPERIENCE

Independent Study: Foundations and RAG with Knowledge Graphs

Aug. 2025 – Present

Arizona State University

Tempe, AZ

- **Analyzed Entity Alignment** techniques for merging heterogeneous graphs, specifically evaluating **Graph Convolutional Networks** and attribute embedding models for cross-lingual integration.
- Evaluated **Retrieval-Augmented Generation (RAG) frameworks**, focusing on how structured knowledge graphs act as a verification layer to mitigate hallucinations in generative systems.
- Explored the **intersection of Linked Data and NLP**, documenting techniques for grounding Large Language Models in factual, curated ontologies to enhance reasoning capabilities.

Volunteer Researcher

July 2025 – Present

Fireline Science

Tempe, AZ

- **Designing and fine-tuning** offline models specifically engineered to assist **Wilderness First Responders** in disconnected areas, addressing critical safety gaps caused by isolation.
- Assisted with the development and testing of Fireline Science's educational workflow system **TWS** including the first use of their external API.
- Implementing **human-in-the-loop** workflows to counteract skill decay and role ambiguity, ensuring reliable guidance for critical care in the wild.

Research Aide and Grader

May 2025 – Present

Arizona State University - W.P. Carey School of Business

Tempe, AZ

- Formulated a **data-driven strategy** for **curriculum optimization** by researching and **implementing LLMs to identify and address organizational skill gaps**, leading to more targeted learning pathways.
- Performed **SEM, ANOVA, ANCOVA, Regression Discontinuity, Survival Analysis**, other statistical and non statistical data analysis, cleaning and visualizations on Canvas LMS Data and other internal data for **industry reports and communicated key findings to sponsors to support strategic decision-making**.
- Facilitate course delivery, **evaluate assignments**, and mentor students in **AWS Cloud Foundations course - CIS 194 (Dept. of Information Systems)**.

MANUSCRIPTS (WORK IN PROGRESS)

Knowledge Graphs & Ontology Alignment | PI: Dr. Srividya Bansal

- DRAL-OA: Results for OAEI 2025 | [Link](#)
- Knowledge Graph Interoperability and Reasoning: A survey of alignment, and RAG
- GEMR:KG – Global Emerging Market Risk Knowledge Graph

AI in Education & Learner Heterogeneity | PIs: Dr. Raghu Santanam, Dr. Collin Sellman

- Parsing Resumes at Scale with LLMs and Predicting Result-Oriented Curriculum for Students
- Individuals in the Dataset: Multi-Modal AI Approaches to Understand Learner Heterogeneity

RESEARCH AND PROJECTS

DRAL-OA | *Researcher*

- Enhanced and submitted a **deep reinforcement and adaptive learning ontology alignment framework** to the **OAEI 2025 campaign**, achieving benchmark scores of **82 percent Precision and Recall** on the complex **anatomy track**. **Currently leading research into integrating MCP and local agentic reasoning model of LLMs** to further **boost alignment accuracy**, with the objective of surpassing the benchmark.

GEMR:KG - Global Emerging Market Risk Knowledge Graph | *Researcher*

- Architected a **neuro-symbolic knowledge graph** merging **World Bank economic and credit risk and political risk datasets** into a unified RDF/OWL ontology on **GraphDB** to model contagion dynamics and temporal correlations. Built a Python-based ETL pipeline for semantic normalization and integrated an **LLM-to-SPARQL interface**, enabling complex multi-hop reasoning to uncover hidden macroeconomic stress signals and geographic default propagation paths across siloed financial data.

Smart Portfolio Builder | *AWS Cloud, Backend Developer*

- Won the MOST INNOVATIVE PROJECT at INNOVATION HACKS - 2025**. Integrated **AWS Bedrock** to provide **AI-enhanced resume summarization** and orchestrated **processing of user portfolio data**. Designed a **Node.js/Express.js RESTful backend**, processing **100+ resumes and videos** via Multer-powered uploads, achieving high uptime through **PostgreSQL**.

HR Onboarding Chatbot | *Full Stack Developer*

- Engineered a **LLM chatbot solution** integrated with **OpenAI modules, sentence transformers, FAISS, NLTK, NumPy, and Pandas**. Built **natural language queries against proprietary content**. Ensured seamless integration with existing **AWS infrastructure**. Pioneered NLP pipeline improving **content retrieval accuracy by 30 percent** and reducing query processing time by 40 percent on AWS.

PROFESSIONAL EXPERIENCE

AWS Cloud and Full-Stack Developer Intern

Aug. 2023 – Jan. 2024

The India Cloud

Bengaluru, India

- Enhanced website **performance and scalability using AWS (EC2, S3, Route 53, CloudFront)**. Established **CI/CD pipelines via AWS CodeBuild, CodeCommit, and CodePipeline** for continuous delivery.
- Built full-stack solutions with **Node.js, React, and Passport** for seamless frontend-backend integration.
- Designed an **AI-driven chatbot** enabling new hires to easily **query HR documents** and ask questions during the onboarding process.

Data Science and AI Internship Training

March 2022 – June 2022

New Age Solutions Technologies (NASTECH)

Bengaluru, India

- Applied AI and data science techniques, including machine learning, deep learning, and big data analysis. Utilized **TensorFlow, Keras, Scikit-learn, Pandas, Matplotlib, and Seaborn** for deep learning model development and visualization.
- Created and deployed a **real-time object-detection service by YOLOv5 on edge devices**, achieving 60 FPS inference (under 50 ms latency).

VOLUNTEERING

Chair of Academia

Jan 2025 – March 2025

ACM at ASU

Tempe, AZ

- Curated a series of **interactive workshops and information sessions** for the **ACM SIG's and Student Research Competition**, guiding **100+ students** through the submission process and enhancing application quality.

Volunteer Teacher

Aug 2020 – April 2024

Noble Cause Foundation

Bengaluru, India

- Tutored underprivileged students** in **STEM** math, science, and social studies.
- Participated in **community outreach events** and cleaning drives, contributing to social welfare.

REFERENCES

Raghu Santanam

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Srividya Bansal

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